

FINESST Ancillary Documents — Drafts

Three short required package items in one working file: split into separate PDFs at submission time per the solicitation's upload structure. [Brackets] = facts to confirm.

A. Facilities, Equipment, and Resources Statement

(No page limit; half a page is typical. Non-anonymized is acceptable here — confirm against the final F.5 text.)

The proposed work requires modest, already-available resources; no facility development or shared-instrument time is needed.

Computing. A workstation with a single consumer GPU ([model], [VRAM] GB) and [N] TB of local storage, owned by [the research group / the FI], is sufficient for all model training and elevation processing in the proposal. The institution provides [institutional HPC name, if any] as surplus capacity; the project does not depend on it.

Software. The full processing stack is open source (Python scientific stack, PDAL, GDAL/OGR, WhiteboxTools, PyTorch, QGIS). No commercial licenses are required.

Data. All required datasets are free and public (proposal \$4 table); the single author-gated item (training labels) has a budgeted fallback. No data purchases.

Workspace. The FI has [office/lab space] in [department], with institutional backup storage for working data and the public archives (Zenodo, EDI) for released products.

Field work. None proposed. The project is entirely archival-data-based; no Antarctic deployment, logistics support, or permitting is required.

B. Acknowledgements (150-word limit, with AI disclosure)

(Required statement identifying the FI as primary author and disclosing all contributions, including AI tools. Count words before submission — target ≤ 150 .)

Draft:

The Future Investigator conceived the proposed research and is the primary author of this proposal. The Principal Investigator provided scientific mentorship and editorial review. AI assistance (Anthropic's Claude, used as a drafting and editing tool under the FI's direction) contributed to text drafting, document formatting, and data-availability verification; all scientific content, claims, and decisions are the FI's own and were reviewed by the FI. Publicly archived datasets consulted in preparing this proposal are credited to their providers: NASA ATM (via USGS), NCALM/OpenTopography, the Polar Geospatial Center (REMA), the McMurdo Dry Valleys LTER program (via the Environmental Data Initiative), Copernicus/ECMWF (ERA5), and NCAR (AMPS). [Add any colleague who gave substantive feedback, including the dissertation author if her input shaped the plan.]

(~120 words as drafted — room for additions.)

C. Budget Justification / Narrative skeleton

(Typically ~2 pages; usually finalized by the PI and the institution's research office — this skeleton gives them the project-specific content. FINESST budgets are capped at

~\$50,000/year total; the composition below is the standard shape — confirm current caps and allowable categories against the F.5 text and institutional rates.)

Year 1 (and similarly Years 2–3):

Item	Amount	Justification
FI stipend	\$[per institutional rate]	[X] months graduate research assistantship — the FI performs all proposed research
Tuition & fees	\$[per institutional rate]	Required for full-time enrollment, a FINESST eligibility condition
Conference travel	\$[~2,500–4,000]	Yr 1: AGU (poster, cryosphere section). Yr 2–3: AGU talk + SCAR/ISAES — the venues named in the Mentoring Plan
Publication costs	\$[~2,000–3,500]	Open-access fees for [1] paper/year (three first-author papers planned)
Computing/storage	\$[~500–1,500]	Local storage expansion for full-valley raster stacks (tens of GB per epoch-derivative); no cloud compute required
Materials/other	\$[small]	[Software-adjacent costs if any; otherwise omit]

Notes for the grants office:

- No PI salary (FINESST does not fund the PI).
- No field-work or logistics costs (archival-data project).
- Indirect costs per the solicitation's rules for FINESST [confirm current F&A treatment in the F.5 text — it differs from standard research awards].
- Year-over-year totals should track the stipend/tuition escalation schedule; the travel/publication mix shifts from poster (Yr 1) to talks + synthesis-paper fees (Yr 3).